

Prelim Exam: The Inflation Deception

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This report examines the historical trends in the US gasoline prices from 1976 to 2025, comparing nominal price movements with inflation-adjusted values to show us the true cost of gas over nearly five decades. We integrated data from the U.S. Bureau of Labor Statistics (Consumer Price Index) and Macrotrends (Nominal Gas Price) which we calculated and adjusted the nominal price using this formula **Real Price = (Nominal Price / CPI) × Base Year CPI**, this dashboard and study shows insights into how inflation has shaped consumer purchasing power and is designed for economists, government officials, energy industry professionals and consumers .

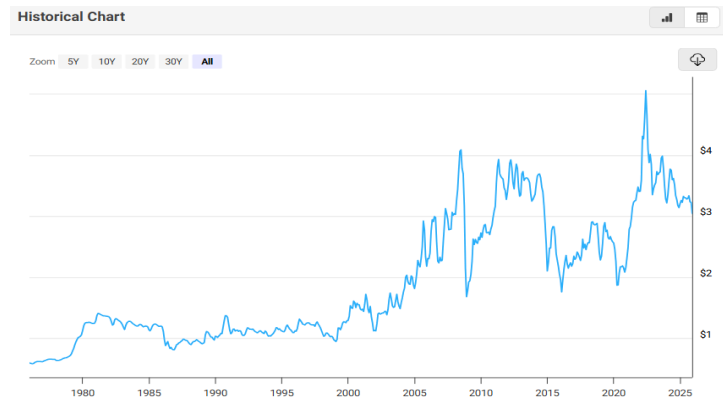
The analysis reveals a significant difference between the nominal and real gas prices. While nominal prices show a clear upward trajectory, inflation-adjusted prices show unpredictability with peak prices in the early 1980s and between 2008-2014 which is then followed by declines that the nominal figures fail to capture.

Towards the color and typography used in this dashboard, The combination of dark background, blue palette for the charts and clean typography created a cohesive visual system to prioritize the emphasis on the data, high contrast ratios were used to ensure the readability and simplifying the dashboard were prioritized to reduce cognitive load which allows viewers to focus on insights rather than decoding the interface.

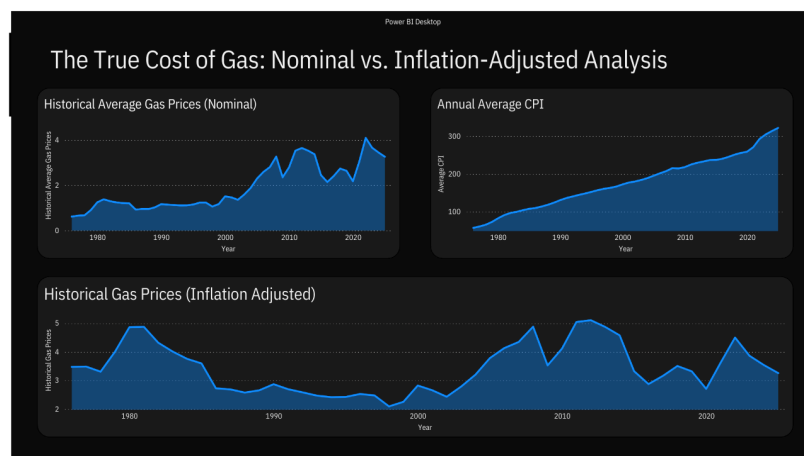
A line chart was selected for all visualizations in the dashboard because it clearly emphasizes the impact of inflation adjustment over time. The top row is intentionally split, placing nominal gas prices alongside the annual average CPI to present the two components that drive the inflation-adjusted calculation. This arrangement guides the viewer's eye from the nominal trend to the inflation factor, before culminating in the bottom, full-width inflation-adjusted price chart, which is given visual prominence as the analytical "punchline." Together, this visual flow highlights how historical gas prices change once inflation is accounted for.

Tufte & Gestalt Audit

Before:



After:



The previous chart only shows the changes in historical gas prices, however it had the following problems:

1. The x-axis only has labels for specific periods every five years, but it still shows extra detail on unlabeled years making the chart feel cluttered.
2. The multiple peaks and lows of the line chart that could divert attention over key points.
3. The light blue lines of the graph is hard to read against a white background.

To fix these issues the following has been done on the redesigned chart:

1. The number of labels has been reduced to simplify the chart and reduce cognitive load.
2. The chart has been smoothed out to highlight key price points on their peaks and lows.
3. The background has been changed from white to dark grey to increase legibility.
4. Additional charts that look simple have been added to add context to the historical gas prices.